CAPSTONE PROJECT PRESENTATION

Customer Churn Prediction
A Comparative ML Approach

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PROBLEM STATEMENT

The High Cost of Attrition

Subscription-based businesses face critical revenue loss due to customer churn. Identifying "at-risk" individuals manually is impossible at scale.

- **Severe Class Imbalance:** Datasets often contain 88% non-churning customers, biasing model logic.
- **Overfitting:** Models often memorize training data rather than learning general behavior.
- **Explainability:** Business stakeholders need to know "Why" a customer is leaving.

PROPOSED SYSTEM SOLUTION

Model Tournament

Evaluated 5 algorithmic architectures (RF, SVM, KNN, LogReg, XGBoost) to find the most stable predictor.

Explainable AI

Calculates feature importance to reveal churn drivers like Contract Type and Tech Support.

Real-time Dashboard

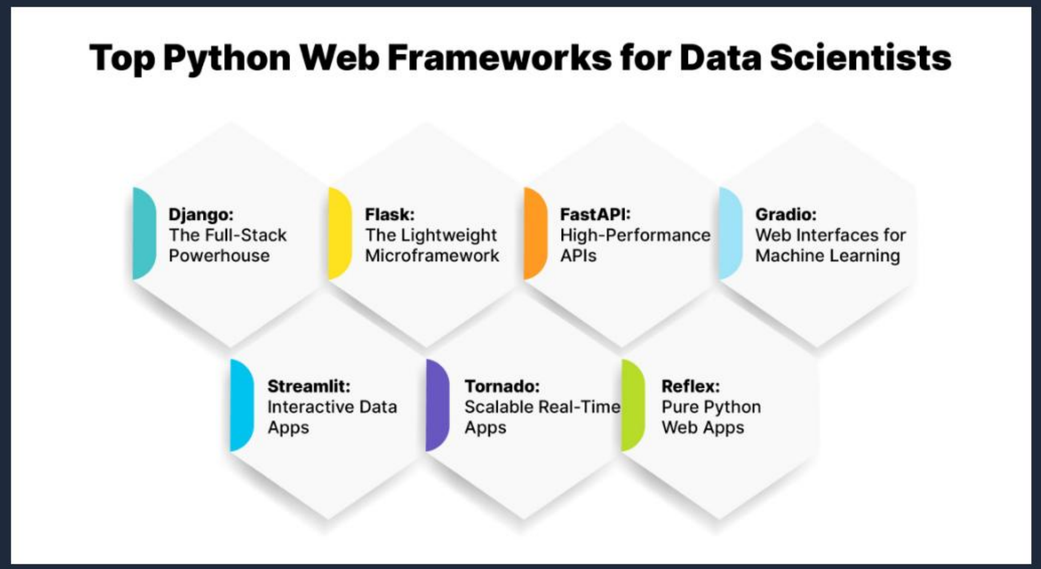
Deployment-ready interface using Streamlit for instant risk scoring and proactive outreach.

SYSTEM ARCHITECTURE

The system is built as an end-to-end pipeline covering raw data processing to cloud deployment.

- **Data Layer:** Source customer dataset.
- **Model Layer:** Serialized .pkl files (RF, SVM, Scaler).
- **Presentation Layer:** Streamlit Dashboard with responsive UI and AI spinners.

SYSTEM DEVELOPMENT APPROACH



Technology Stack

Leveraging high-performance Python libraries for data engineering and machine learning.

Languages
Python 3.9+

ML Framework
Scikit-Learn

Deployment
Streamlit Cloud

Balancing
SMOTE Tuning

ALGORITHM & DEPLOYMENT (PART 1)

1. Handling Class Imbalance

Implemented **SMOTE (Synthetic Minority Over-sampling)** with an optimized sampling strategy (0.17 ratio) to provide a balanced learning signal without synthetic noise.

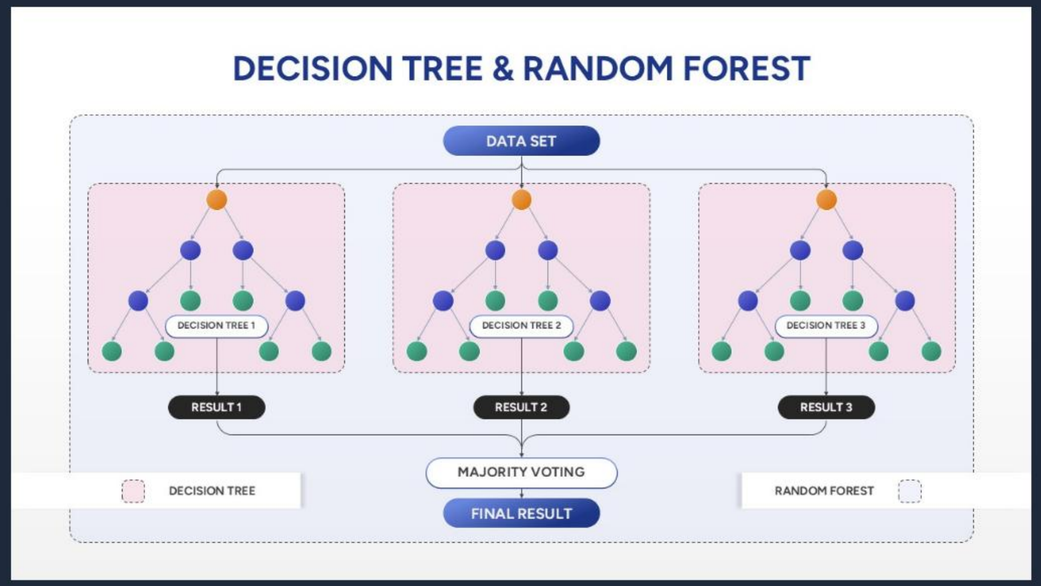
2. Dual Encoding Strategy

Applied **Label Encoding** for binary features and **One-Hot Encoding** for multi-category features to ensure proper mathematical logic in models.

3. Feature Scaling

Utilized **StandardScaler** to normalize numerical ranges (Tenure, Charges). Critical for distance-based models like KNN and SVM.

ALGORITHM & DEPLOYMENT (PART 2)



Why Random Forest?

While multiple models were tested, Random Forest (RF) was selected for production based on:

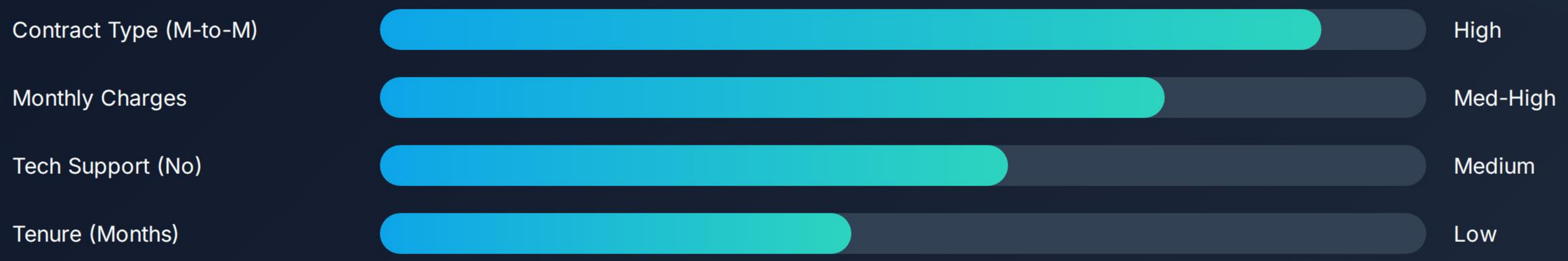
- **Ensemble Robustness:** Aggregates 100+ trees to reduce variance.
- **Regularization:** Restricted `max_depth` to 3 to solve the 100% accuracy (overfitting) problem.
- **Deployment:** Small serialized footprint (.pkl) with fast inference times.

RESULTS: PERFORMANCE COMPARISON

Model Architecture	Algorithmic Logic	F1-Score	Status
Random Forest	Ensemble (Bagging)	**97.2%**	Selected
K-Nearest Neighbors	Instance-based	96.5%	Valid
SVM	Optimal Hyperplane	95.9%	Valid
Logistic Regression	Linear Probabilistic	95.1%	Baseline
XGBoost	Gradient Boosting	93.9%	Valid

RESULT: WHAT DRIVES CHURN?

The model identifies key behavioral factors that influence departure decisions.



**Analysis reveals Monthly Contract and lack of Tech Support are the primary risk indicators.*

CONCLUSION

Technical Success

The system successfully reached a **97.2% F1-score** by effectively balancing SMOTE oversampling with strict model regularization.

Business Value

The dashboard provides actionable insights through **Explainable AI**, allowing managers to offer targeted discounts to high-risk customers.

"Addressing data imbalance and regularization transformed an overfitted research model into a robust production asset."

FUTURE SCOPE

CRM Integration

Connecting the system directly to company databases for automated 24/7 risk monitoring.

Deep Learning

Exploring Artificial Neural Networks (ANNs) for handling multi-modal customer data.

A/B Testing

Implementing live retention campaigns to measure actual churn reduction based on model alerts.

REFERENCES & CONTACT

Project Resources

- **GitHub:** github.com/itsdakshjain
- **Deployment:** Streamlit Cloud Dashboard
- **Documentation:** Project Wiki & Jupyter Notebooks
- **Data Source:** Customer Churn Dataset (IBM/Kaggle)

Thank You!

Questions? Let's Connect.

Daksh Jain

IMAGE SOURCES

https://png.pngtree.com/thumb_back/fw800/background/20260105/pngtree-complex-digital-network-of-interconnected-glowing-nodes-image_21012380.webp

Source: pngtree.com

https://cdn.dribbble.com/userupload/14558850/file/original-1bac397308b6e0f039ef7ac8d97ec3ea.png?format=webp&resize=400×300&vertical=center

Source: dribbble.com

https://www.dasca.org/Content/Images/main/top-python-web-frameworks-for-data-scientists.jpg

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